

Basic Principles of Medicine 2
Module: Endocrinology and Reproduction.
DLA Video

Infant nutrition

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Recommended Reading

Lippincott Illustrated Reviews (Lippincott) 8th edition

Pages 441 and 450

<https://meded.lwwhealthlibrary.com/book.aspx?bookid=3073>

<https://www.cdc.gov/infant-toddler-nutrition/index.html>

Objectives: DLA: Nutritional Aspects of Lactation and Infant Nutrition

- DLA to be reviewed before “Nutritional Aspects of Pregnancy, Lactation and Infant Nutrition”

SOM.MK.III.BPM2.1.ER.2.BCHM.1055	Highlight the major differences in the composition of colostrum, transition milk and mature milk during lactation.
SOM.MK.III.BPM2.1.ER.2.BCHM.1056	Discuss the advantages of human milk compared to animal milk and infant formula.
SOM.MK.III.BPM2.1.ER.2.BCHM.1057	Review WHO guidelines about breastfeeding and HIV.
SOM.MK.III.BPM2.1.ER.2.BCHM.1058	Identify the age range of neonate and infant. Discuss Vitamin K deficiency bleeding (VKDB) in the neonate and infant.
SOM.MK.III.BPM2.1.ER.2.BCHM.1059	Identify micronutrients that are low in human breast milk. Highlight the dietary modifications (iron) to be implemented after 6 months of age.
SOM.MK.III.BPM2.1.ER.2.BCHM.1060	Describe bioavailability of iron and factors that reduce inorganic iron uptake.

Infant Nutrition

- Good nutrition during the first 2 years of life is vital for healthy growth and development
- Early start help develop healthy dietary patterns
- Breast milk is a great source of nutrition for babies
- Infants and children are in a state of positive nitrogen balance (growth)

Note: Always adhere to your country's/region's recommended nutritional guidelines to ensure safe, evidence-based care for every patient.

World Health Organization (WHO) recommendation

Exclusive Breastfeeding

- Exclusively breastmilk for the first 6 months.
- No other foods or liquids, including water, should be given during this period.

Initiation of Breastfeeding

- Begin within the first hour after birth

Continued Breastfeeding

- After 6 months, continue breastfeeding plus introduction of nutritionally adequate complementary foods

On-Demand Feeding

- Breastfeed whenever signs of hunger day or night

Avoiding Bottles and Pacifiers to promote successful breastfeeding



Age range of neonate and infant

- **Neonate**: Birth to 28 days of life
- **Infant**: Birth up to 12 months of age
 - **Early Neonatal**: <7 days
 - **Late Neonatal**: 7–27 days
 - **Post-Neonatal**: 28 days–1 year



Clinical case

Patient Information

- **Name:** Timmy the "Tiny Titan"
- **Age:** 2.5 months
- **Sex:** Male

Chief Complaint: Mother reports "Vomiting and loose stools for 2 days."

History of Presenting Illness

- Onset: **2 days ago**
- Symptoms: **Frequent vomiting** and **loose watery stools**
- No reported fever, blood in stools, or respiratory symptoms
- Feeding during illness: **Continued formula feeding and glucose drink**

Past Medical History: Unremarkable

Growth history:

- At **1 month:** Weight at 50th percentile
- At **2.5 months:** **Weight now <3rd percentile** → **significant growth faltering**



Nutrition History

- **Breastfeeding:** Exclusive up to 1 month
- **Formula feeding:**
 - Dutch Lady 1 and Lactogen 1
 - Dilution: **2 scoops per 3 oz water** (appears correct per standard instructions)
- **Additional feeds:**
 - Glucose drink (1 scoop per 2 oz water)
 - Plain porridge (2 tablespoons twice daily)



Clinical Findings

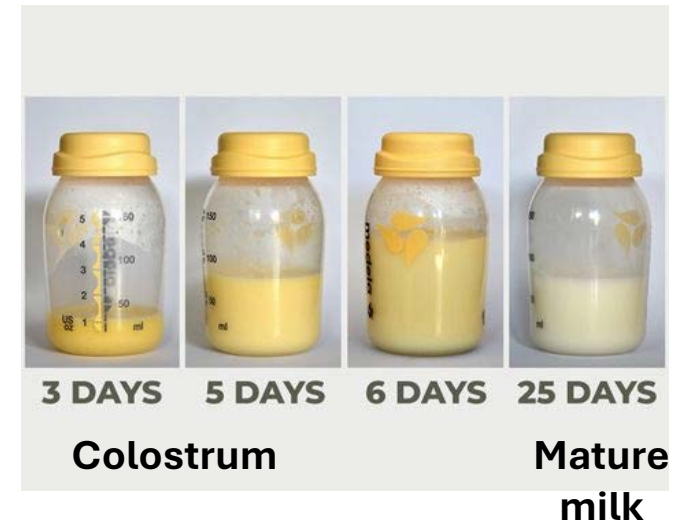
- **General appearance:** Underweight, alert but irritable
- **Hydration status:**
 - Dry lips, slightly sunken eyes, reduced urine output
 - **Mild dehydration suspected**
- **Vital signs:** HR 160 bpm → dehydration causes mild tachycardia, RR 55 breaths/minute → mild increase due to compensatory response
- **Anthropometric Data:**
 - Weight: **3.25 kg** (weight-for-age: <3rd percentile WHO growth chart)
 - Length: **57 cm** (length-for-age: ≈50th percentile)
 - Weight trajectory: Dropped from **50th percentile at 1 month** to **<3rd percentile now**
 - Weight-for-length: **severely wasted**

Stages of Breast milk

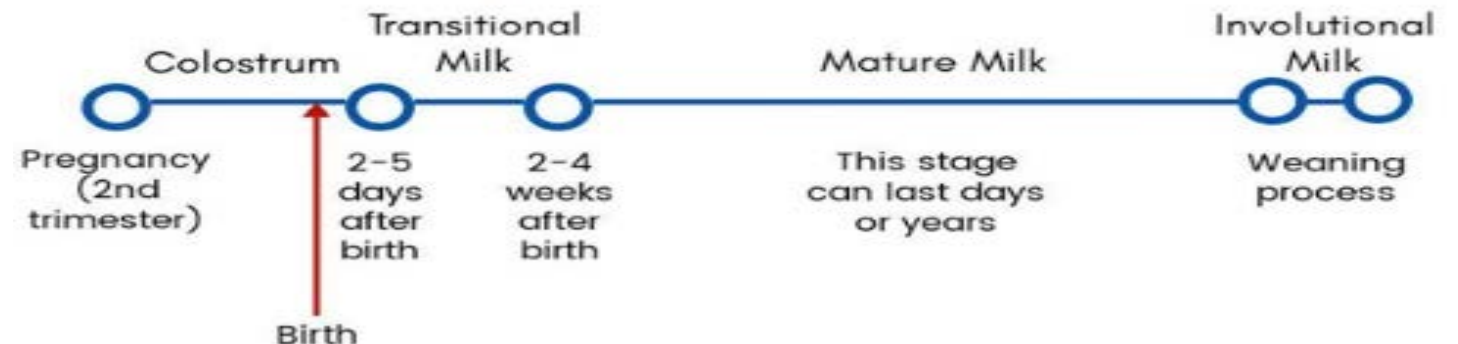
- Milk is a dispersion of:
 - ✓ milk fat globules with TAGs
 - ✓ casein micelles with protein, calcium and phosphorus

- **Colostrum**

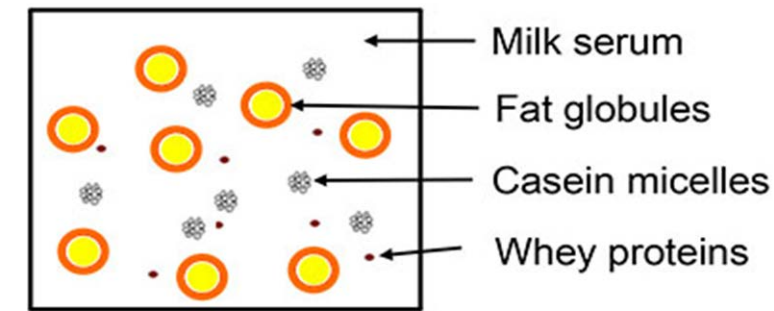
- First milk produced during late pregnancy until 3-4 days postpartum
- Thick, yellowish substance
- Rich in proteins and immunoglobulins (**IgA**)
- Low in lipids and carbohydrates



Stages of Breast Milk



Components of Mature Breast Milk



- Phase of milk serum with lactose, whey protein, vitamins and minerals

Carbohydrate	Lactose and prebiotic human milk oligosaccharides (HMO)
Fat	TAGs (triglycerides) and cholesterol Fatty acids in TAGs: Medium- and long chain fatty acids synthesized in lactating mammary gland. Also rich in oleic acid (36%) and PUFA ($\omega 6 > \omega 3$) from maternal diet including arachidonic acid (ARA) and docosahexaenoic acid (DHA)
Protein	High whey to casein ratio in comparison to cow's milk Nutrients rich in essential and conditional essential amino acids Proteins with non-nutritive functions
Vitamins	Rich in vitamin A , B-vitamins and vitamin C Low in vitamin K and vitamin D, sunlight exposure of infant recommended
Minerals	High in sodium, potassium, magnesium, zinc, chloride Low in iron, calcium and phosphorus (but highly absorbed)

Types of proteins in milk

1. Whey

- ✓ Mostly α -lactalbumin
- ✓ Easier to digest than casein. It does not precipitate at acid pH.
- ✓ Rich in leucine for stimulation of protein synthesis.

2. Casein

- ✓ Casein micelles contain water, calcium and phosphorus
- ✓ Slow digestion of the protein, precipitates at acid pH (curd)
- ✓ Rich in proline for collagen synthesis

Whey/casein ratio changes: highest in colostrum

Advantages of whey in human milk

α -lactalbumin

- predominant protein in whey (whey protein)
- rich in cysteine and tryptophan (helps with the positive nitrogen balance)

Lactoferrin:

- binds iron and inhibits growth of destructive iron-dependent bacteria in the GI tract of neonates

Lysozyme

- Enzyme that protects infant against E. coli and Salmonella
- promotes the growth of healthy intestinal flora
- has anti-inflammatory functions

Immunoglobulins:

- mostly IgA which inhibits binding of pathogens
- protects the immature intestine of babies

Human Milk compared to Cow milk and Infant Formula

	Breast Milk	Cow Milk	Formula Milk
Water	High	Low	May need extra
Protein ★	Low	High	Partially corrected
Carbohydrates ★	Lactose and probiotic Human milk oligosaccharides (HMO)	Lactose- less oligosaccharides	Lactose + Sucrose Lacks lipase
Fat	EFAs, Lipase	No EFAs and lipase	Some EFAs, no lipase
Vitamins ★	Adequate depending on mom's nutritional adequacy (A,B,C)	Low vitamin A, C and Iron	Vitamin/mineral added- usually enough
Immunoglobulins ★	Ig A, Lactoferrin, Lysozyme, etc.	None	None
Whey/Casein ratio ★	High	Low	Depends on formulation
Growth factors	Present	None	None

Possible contributing factors

1. Inappropriate additional feeds: glucose drink → empty calories, risk of diarrhea
2. Early cessation of breastfeeding
3. Potential formula intolerance or infection



Management Flowchart

1. Assess hydration → Oral rehydration if mild, IV fluids if severe
2. Stop inappropriate feeds (glucose drink) → worsens diarrhea
3. Correct formula preparation → Standard: 1 scoop per 30 mL water
4. Consider lactose-free formula if diarrhea persists
5. Monitor growth and hydration
6. Educate caregiver on feeding practices

Types of infant formula



1. Cow milk-based formula (often whey added)
2. Soy-based formula (not for premature infants)
3. Specialized formula (hypoallergenic or lactose-free)

Additives may include:

- Lactoferrin, HMO, ARA, DHA, lutein, taurine, choline, vitamins, and many other components dependent on the chosen formula.



HIV and Breastfeeding Recommendations



- **CDC**
 - ✓ breastfeeding may be considered if they are on ART with sustained viral suppression
 - ✓ shared decision-making counseling
 - ✓ close monitoring
- **American Academy of Pediatrics (AAP)**
 - ✓ breastfeeding can be supported for mothers on ART with undetectable viral load
 - ✓ but formula or donor milk is recommended for those not virally suppressed
- **WHO**
 - ✓ exclusively breastfeed for six months
 - ✓ continue breastfeeding with complementary foods for at least 12 months while on lifelong ART



Neonatal hemorrhagic disease

Vitamin K Deficiency Bleeding (VKDB)



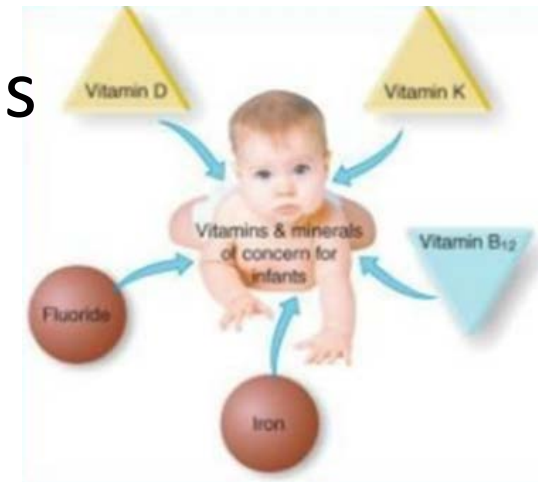
- **Maybe within** 24 hours, 1 week or between 2-8 months after birth
- Insufficient vitamin K affects maturation of blood clotting factors (II, VII, IX and X)
- **Clinical features:** Easy bruising, hematuria, intracranial, and gastrointestinal **hemorrhages** (due to deficiency of mature clotting factors)
- Factors contributing to vitamin K deficiency in infants:
 1. Poor placental passage of vitamin K
 2. Underdeveloped intestinal flora (which produces vitamin K) esp in premature infants
 3. Low vitamin K levels in breast milk
 4. Vitamin K1 is rich in green vegetables which are not part of the neonatal diet.
 5. Low liver storage capacity



Treatment: Prophylactic intramuscular **vitamin K administration** at birth (0.5-1mg)

Dietary modification for infants older than **6 months**

- Breast milk provides most essential nutrients
- Some micronutrients are present in low concentrations
 - **Iron**: highly bioavailable but insufficient in quantity
 - **Vitamin D**: often inadequate, needing supplementation
 - **Vitamin K**: low requiring prophylaxis to prevent bleeding
 - **Vitamin B12**: Especially if maternal diet is vegan.



Iron supplementation after 6 months of age

During pregnancy

- mother provides iron for an increased iron depot to the fetal liver
- supplies the baby up to **six months** after birth

Six (6)-month-old baby

- Human milk is low in iron (0.35 mg/L)
- Infants need about 11 mg of iron/ day (cannot be provided by milk alone)
- Include iron-rich foods

Recommendations for iron



Iron-Rich Complementary Foods

- Introduce pureed meats, poultry, and fish as primary heme iron sources from six months
- Heme iron better absorption

Iron-Fortified and Plant Sources

- Iron-fortified cereals and plant-based sources (lentils) paired with vitamin C foods
- Vitamin C enhances absorption of non-heme iron

Avoid Cow's Milk Early

- Avoid giving cow's milk before 12 months
- Inhibits iron absorption and increases anemia risk

Feeding Strategies: Combining iron-rich foods with vitamin C and avoiding inhibitors supports optimal iron uptake in infants

Pureed food rich in vitamins and minerals

- **Avocado**: Rich in essential fatty acids and minerals
- **Banana**: Electrolytes and calcium
- **Carrots, peas, green beans, broccoli**: β -carotene and minerals
- **Squashes, sweet potato**: Easy to digest carbohydrates
- **Pears**: Least acid fruit, Vit K and copper



food	6 months	7-9 months	8-12 months
Breast Milk or Formula	4-5 feedings	3-5 feedings	3-4 feedings
Cereal/ Grain	2 meals	2 meals	2 meals
Fruits	2 meals	2 meals	2 meals
Vegetables	2 meals	2 meals	2 meals
Fruit Juice (optional)		1 serving	1 serving
Protein Foods		1 meal	2 meals

* A meal is approximately 2 to 4 tbsp

Summary

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- Breast feeding recommendations
- Stages of breast milk and composition
- Comparison between breast milk, cow's milk, and formula
 - Protein content; whey/casein ratio
 - Lactose and HMO
 - IgA and lysozyme
- Vitamin K prophylaxis
- Dietary modifications infants >6 months
 - Iron, vitamins D, K, and B12
- **Reminder: Always adhere to your country's/region's recommended nutritional guidelines to ensure safe, evidence-based care for every patient.**